

David Vasilev

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EDUCATION

Purdue University - West Lafayette
B.S. Aeronautical and Astronautical Engineering
• Purdue Solar Racing Team: Chief Engineer

Expected Class of 2027
Dean's List | 3.66/4.00

Relevant Coursework:

- Aeromechanics, Structural Analysis, Thermodynamics, Aerodynamics, Fluid Mechanics, Dynamics and Vibrations, Control Systems and Signals Analysis, Object-Oriented Programming, Data Structures

EXPERIENCE

General Atomics - Aeronautical | Avionics Mechanical Engineer

San Diego, CA

General Atomics

Jun. 2026 – Aug. 2026

- Engineered custom metal enclosures, mounting brackets, and structural chassis for PCBs, collaborating with electrical design and system integration teams for optimal packaging and performance.
- Executed thermal simulations to validate avionics component survivability, optimizing heat dissipation paths and ensuring long-term hardware reliability across environmental operating conditions.
- Produced detailed 2D manufacturing drawings applying ASME Y14.5 Geometric Dimensioning and Tolerancing standards to guarantee precise fabrication and seamless assembly.
- Optimized ECAD-to-MCAD data translation workflows utilizing, accelerating design integration and iterations.

Purdue Solar Racing Club | Chief Engineer

West Lafayette, IN

American Solar Challenge, Formula Sun Grand Prix

Aug. 2024 – Jul. 2026

- Led 200+ member team in top-down subsystem integration for 2026 competition, coordinating schedules, cross-team workflows, and weekly deliverables while dedicating 70+ hours/week.
- Solved complex packaging challenges by designing chassis layouts under tight spatial constraints; optimized component placement and orientation for cooling, wire efficiency, and structural integration.

Purdue Aerial Robotics Team | Avionics and Optics Team Member

West Lafayette, IN

Student Unmanned Aerial Systems Competition

Aug. 2024 – Aug. 2025

- Derived electrical and optical component requirements to meet system specifications for the 2025 competition.
- Performed Finite Element Analysis of mounts under crash load conditions using Siemens Simcenter.
- Conducted thermal and electrical load analysis on batteries and data processing components.
- Designed avionics installation and defined wiring harness specifications, performed dynamic analysis to optimize RF cable lengths, antenna separation, and electrical signal transmission losses.

PROJECTS

Solar Car - Artemis | Top-Down Modeling, System Integration

Aug. 2024 – Jul. 2026

- Applied top-down modeling and WAVE linking in Siemens NX to enable in-place design and full-system integration across mechanical, electrical, and structural assemblies.
- Resolved integration challenges by aligning electrical systems and mechanical assemblies with chassis geometry, optimizing spatial layout, serviceability, and mechanical compatibility across subsystems.
- Managed and validated CAD workflows via PDM; defining structural interfaces for rigidity and aerodynamic fit.

Monocoque Chassis | Composites, CAD, Finite Element Analysis

Aug. 2024 – Jul. 2026

- Designed a pre-preg carbon fiber monocoque chassis as the vehicle's central load-bearing structure, integrating suspension, electronics, and driver systems into a unified geometry.
- Integrated CG targets, suspension hard-points, safety regulations, and packaging constraints into layout; optimized stiffness-to-weight ratio and selective coring via FEA to reduce mass while maintaining crash safety.
- Generated layout-ready geometry and tooling surfaces in Siemens NX to support composite manufacturing and iterative design validation.

SKILLS

Design Tools: Siemens NX/Simcenter/Teamcenter, Creo, Ansys Icepak/Mechanical/Fluent, Altair Hypermesh, Autodesk Inventor/Fusion 360, SolidWorks, OnShape, Bambu Slicer

Manufacturing: Hand Layups: VARTM, Pre-Preg, Wet; ShopeSabre 5000, Bambu X1C/X2D/P1S, AnkerMake M5

Programming Languages: Matlab, Powershell, Python, C, Java, HTML, JS, CSS